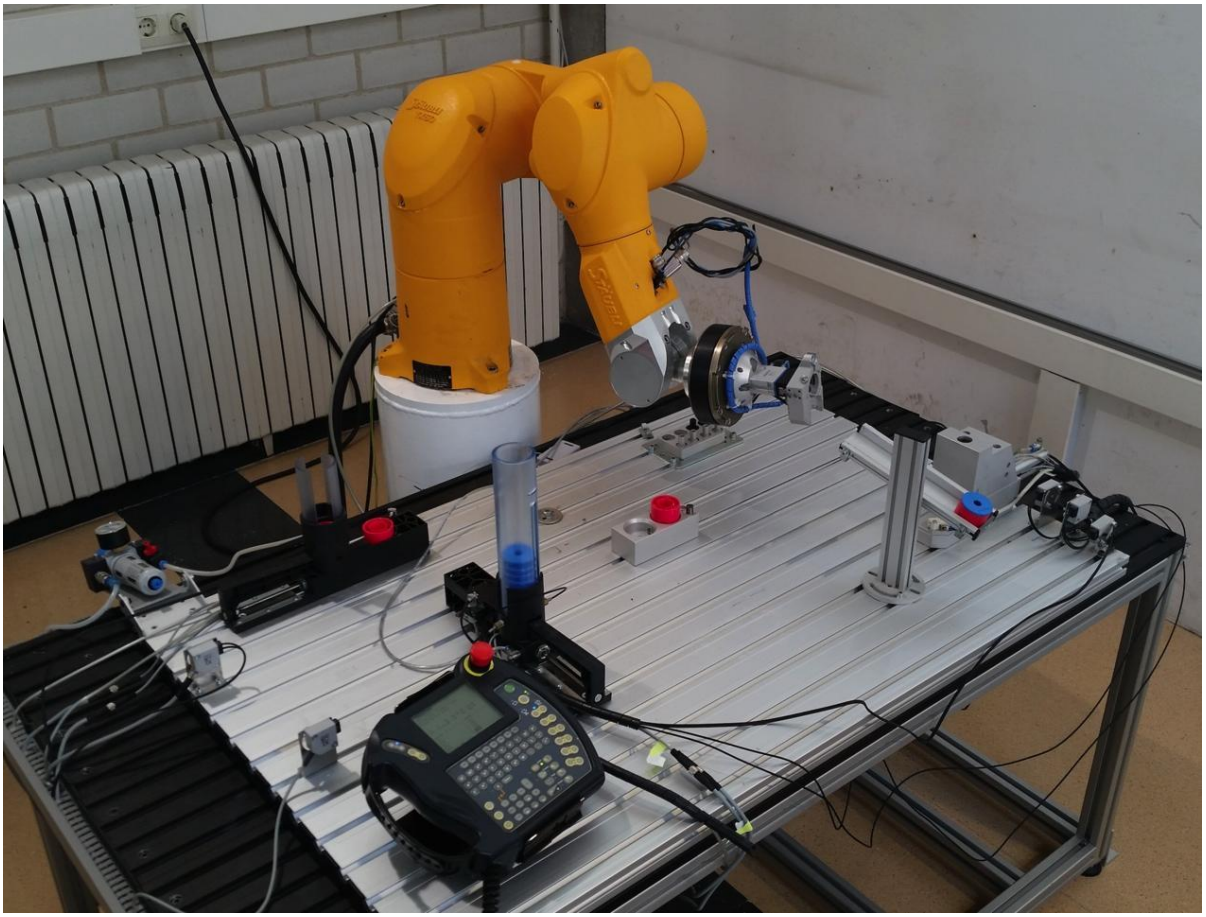
Robot Manipulation

Pick and Place with the Stäubli TX60



This document aims to explain the work to be done in the practice of assembly using the Staubli robot. You can consult the "SRS reference manual" and the "VAL3 reference manual, version 6" when necessary. You need to have completed the SRS "Tutorial", in which you learn how to create an application and simulate it with the emulator. You must have the SRS program installed and use the university license.



IMPORTANT. You should carefully follow the instructions below to avoid any chance of an accident.

- Make sure that in your code the maximum speeds of mdesc type variables are never higher than 30. Never test at a speed higher than 30% of the maximum speed.
- Always lower the speed from the MCP to 10% the first time you try a program. If the program works properly, you can then increase the speed from the MCP to 100% (30% of the maximum speed defined in the code).
- Before pressing the Move / Hold button, locate the STOP and EMERGENCY STOP buttons on the MCP, the external emergency stop switch, and check that the barriers are properly armed.
- Before pressing the Move / Hold button, make sure that no one is in the environment marked by the manipulator's security barriers.
- During execution, use the Move / Hold button to stop and restart execution.
- During execution, use the STOP button to stop the entire application. Be careful, it won't stop immediately, press Move / Hold first.
- Only in case of danger to a person or the manipulator do you press the EMERGENCY STOP buttons.

1. Introduction

This lesson consists in the programming of the Stäubli TX60 to develop a simplified robot manipulation in a Pick and Place environment.

The student will work with the computer and also directly with the robot. Like there is only one robot for all the students, it will be required to work as much as possible in simulation.

The lesson consists of different levels, allowing the students to reach one or another, depending on their skills and interest.

2. Cell description

In this section, the different elements of the cell are presented, see Fig.1.

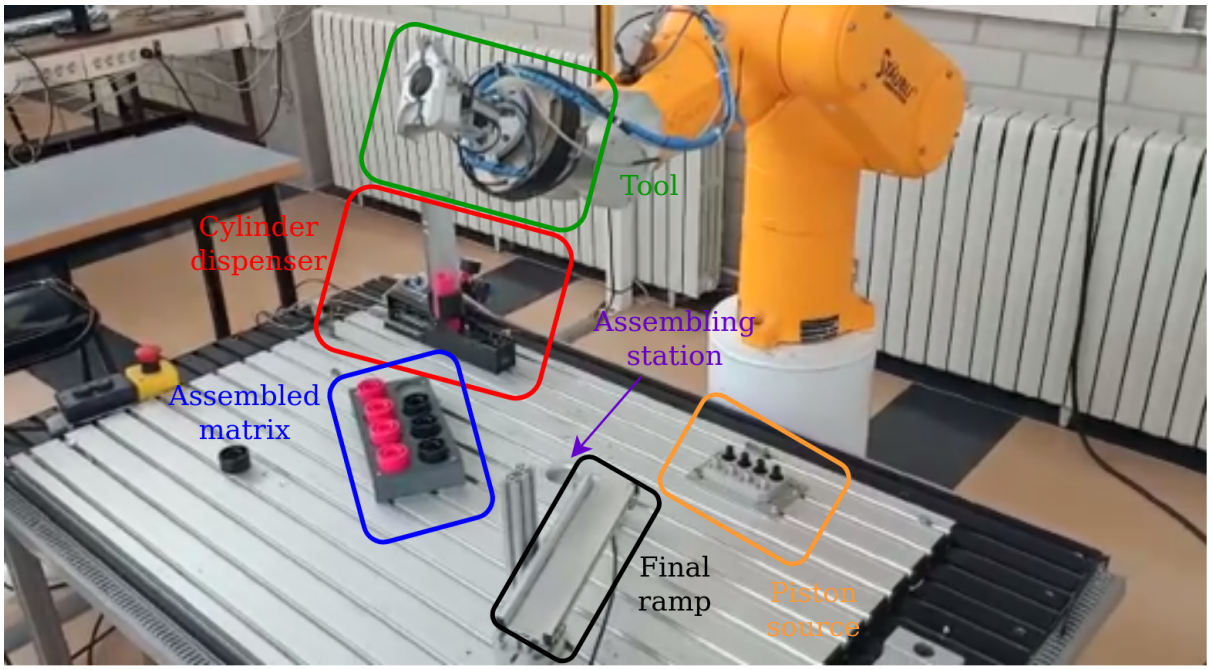


Figure 1: Cell elements

2.1 Parts

In this lesson two parts have to be assembled: the *red cylinder* and the *black piston* (see Fig.2). They are a representation of a real assembly adapted to do not have a huge demand in precision. The *black piston* has to be deposited inside the *red cylinder*.



Figure 2: Red cylinder and black piston

2.2 Tool

The tool is the mechanical element added to the robot to be able to operate with the different parts. In this case, the tool (see Fig.3) is composed of one pneumatic actuator that allows the robot to open and close the gripper with pneumatic power and a digital signal. The gripper has multiple points of grip, designed for each part. **Be extremely cautious operating with the gripper, you can trap your fingers and there is no safety to avoid it**.

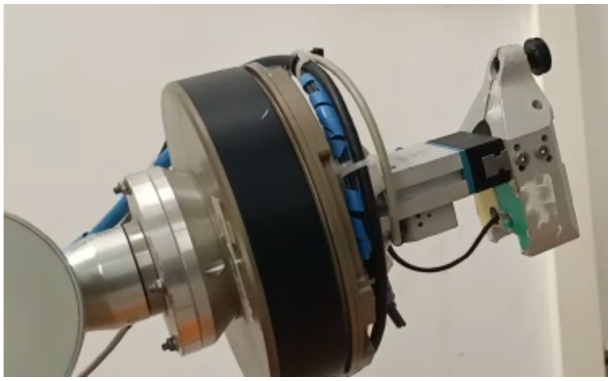


Figure 3: Tool of the robot

The first point of grip is called *Lateral tool*, it is designed for pick the cylinders, and it's configuration is:

x	y	z	rx	ry	rz	gripper	otime	ctime
30	30	170	-90	44	-90	io:valve1[0]	1	1

The second point of grip is called *Tip tool*, it is designed to pick the pistons, and it's configuration is:

x	y	z	rx	ry	rz	gripper	otime	ctime
-30	-30	175	0	0	-134	io:valve1[0]	1	1

2.3 Cylinder dispenser

The *Cylinder dispenser* is a simple structure with a pneumatic actuator that allows the feeding of cylinders at the station, see Fig.4. It has a presence sensor in order to know the presence of the cylinder in the picking place.

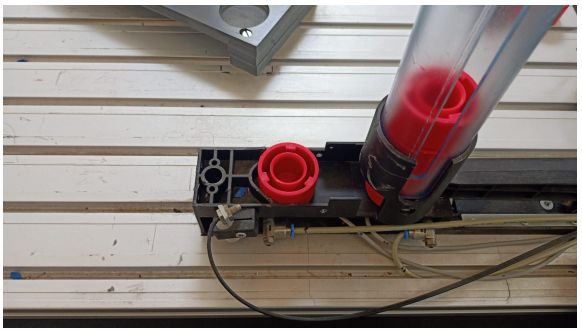


Figure 4: Cylinder dispenser

2.4 Piston source

The *Piston source* is a tray where we manually put the pistons, see Fig.5.



Figure 5: Piston source

2.5 Assembling station

The *Assembling station* is a place prepared to manipulate the parts. It has different positions and a presence sensor, see Fig.6.



Figure 6: Assembling station

2.6 Final ramp

The *Final ramp* is a place where the parts can be placed, see Fig.7.

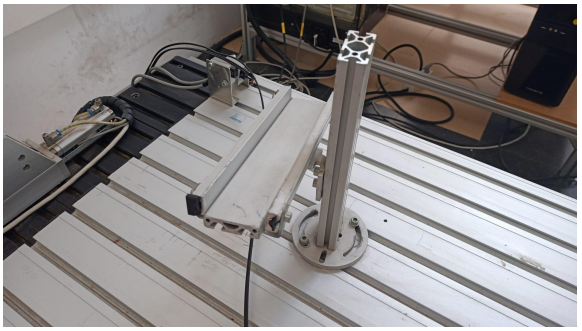


Figure 7: Final ramp

2.7 Assembled matrix

The *Assembled matrix* is a table where there are different positions to place the cylinders, see Fig. 8.



Figure 8: Assembled matrix

It is recommended to create a frame related to this object, in order to define the position in an optimal way.

2.8 Inputs/Outputs map

Signal	Logic	Description
bOut0	On (open) / Off (close)	Output to control the solenoid valve to open and close the gripper of the tool.

bOut10	On (order to extend the pneumatic actuator) / Off (Order to contract the pneumatic actuator)	Control of the solenoid valve to move the pneumatic piston of the *Cylinder dispenser*.
bIn0	On (presence) / Off (no pressence)	Presence sensor of the tool. Sensor to know if there is a *piston* in the *piston source* for Level 3.
bIn2	On (presence) / Off (no presence)	Sensor to know if there is a cylinder ready to be picked in the *Cylinder dispenser*.
bIn14	On (pneumatic actuator extended) / Off (pneumatic actuator not extended)	Sensor to know if the pneumatic actuator of the *Cylinder dispenser* is extended.
bIn15	On (pneumatic actuator contracted) / Off (pneumatic actuator not contracted)	Sensor to know if the pneumatic actuator of the *Cylinder dispenser* is contracted.

3. Memory

The *Pick and place with the Staübli TX60* practical lesson consist of different levels. In each one, the student should program the application on his computer, using SRS. When the application is ready in simulation, the student must show it to the teacher in order to have the acknowledgment to use the robot. When the robot is available and the application is ready, the student can test the program in the real robot.

In the simulation scenario, you have the environment already prepared with the different elements of the laboratory. You can find the file “Entorn_TX60.zip” in the moodle, with the files related to the environment, you must load this file to the SRS.

In the simulation, the student will develop the algorithm required to perform the demands but will put the positions of the robot in a safe place, keeping a conservative distance between the different elements. Would be in the real robot, when the student will adjust the points to the exact required position.

3.1 Level 1: Basic assembly

The robot has to assemble two parts: the *red cylinder* with the *black piston*. It has to pick the cylinder from the *Cylinder dispenser* (see Fig.9) and the piston from the *Piston source* (see

Fig 11). It has to assemble in the *Assembling station* (see Fig.10, Fig. 12) and place the assembled part in the *Final ramp* (see Fig.13).

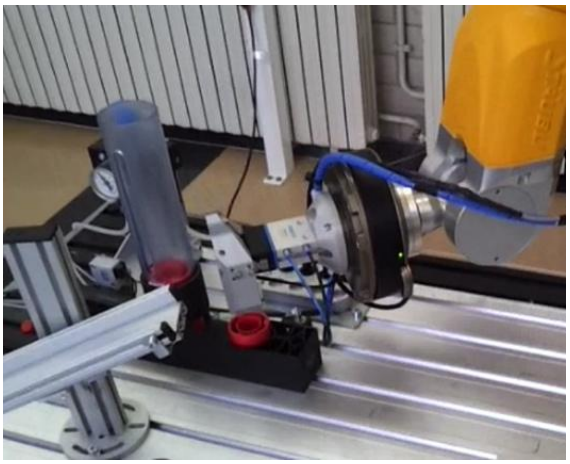


Figure 9: Robot picking the red cylinder from the Cylinder dispenser

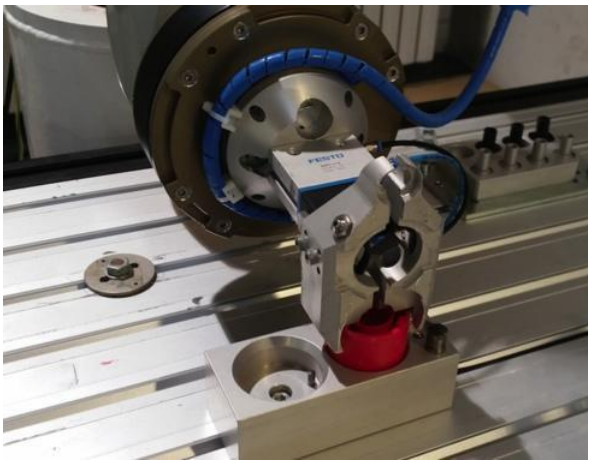


Figure 10: Robot placing the red cylinder to the Assembling station

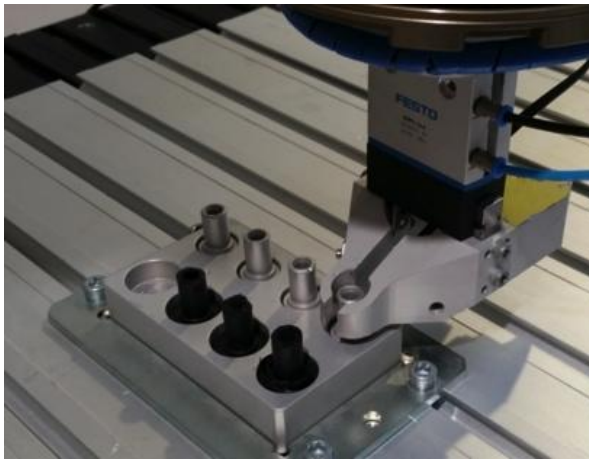


Figure 11: Robot picking the piston from the Piston source



Figure 12: Robot assembling the piston into the cylinder

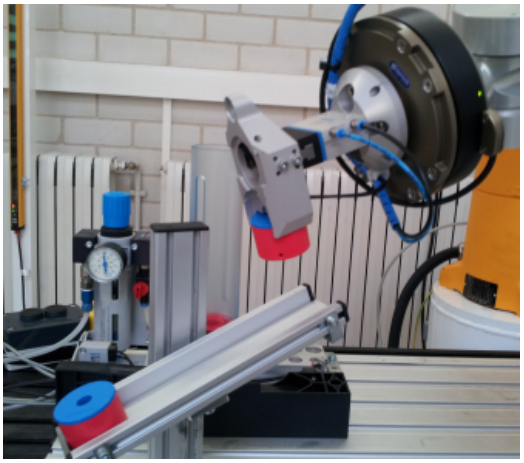


Figure 13: Robot placing the assembly into the Final ramp

3.2 Level 2: Continuous feeding

The robot has to assemble the two parts (*red cylinder* with the *black piston*) in a continuous way. The robot has to take the control of the *Cylinder dispenser* and feed it with *red cylinders* (controlling the pneumatic actuator), pick the *red cylinder* from there, assemble the *black piston* and place the assembly into the *Assembled matrix*. This process has to be repeated four times.

3.3 Level 3: Continuous feeding with troubles

When you program an industrial robot, one of the basic things is to have control of every variable that can be modified in your system. For example, when you make a pick, you have to guarantee that there is a part to be picked.
Trying to emulate this concept, a presence sensor is added to the tool (see Fig.9). This presence sensor detects if there is something in front of it.

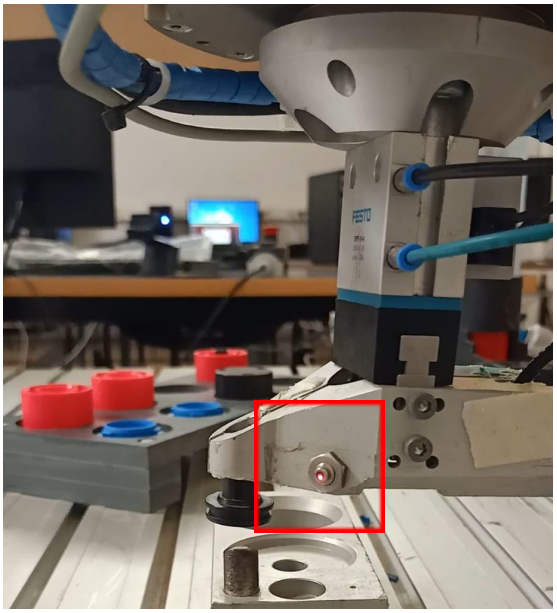


Figure 9: Presence sensor of the tool

At this level the robot has to make the assembly like in Level 2, taking the red cylinder from the *Cylinder dispenser* and taking the piston from the *Piston source*. But, in this case, not all the positions of the *Piston source* would be with a piston, so the robot would have to recognize (using the presence sensor of the tool) if in the position that it wants to pick, have a piston or not and take another one if it's necessary. With the assembly done, it has to place it in the *Assembled matrix*.

4. Evaluation

The students have to present a report explaining the practical lesson and how they develop the code in order to face the different problems. They have to present also a video of each performance of the robot developed.

The evaluation criteria will follow different points:

- Level reached.
- Performance of the robot in each level.
- Use of tools and frames.
- Intuitive messages in the MCP.
- Good structuration of the code, self-explaining code, and usage of the optimal functions.
- Good presentation of the report.